Synthesizability as a Design Constraint: A Physics-Informed Generative Pipeline for Auditable Early-Stage Molecular Design

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Abstract

Structure-based generative models now propose molecules at a rate no medicinal-chemistry team can triage, yet most pipelines apply synthesizability *after* generation, as a filter on a candidate list. We describe a pipeline in which route feasibility is treated as a constraint at the point of design, and in which every prediction carries an explicit epistemic label rather than an implicit score. The pipeline couples target-conditioned de novo generation, structure-based docking with pose-level output, multi-pose physics-based rescoring, independently cross-checked interaction fingerprinting, a liability screen spanning ADMET, cardiac and hepatic endpoints, a CYP450 interaction panel, and per-candidate retrosynthetic route proposal from commercially available starting materials. We report a retrospective pose-recovery case study on the Parkinson's disease kinase LRRK2 against the reference inhibitor MLi-2, and we set out the evaluation protocol — benchmark pose recovery, decoy discrimination, route-feasibility adjudication, and ablation of the design constraints — that a prospective claim would require. We are explicit about what this work does not show: no compound in this study has been synthesized or assayed, and pose recovery is not evidence of prospective discovery. We contribute the architecture, a reporting discipline for in-silico design claims, and a checklist that makes the gap between a ranked docket and a validated hit auditable.

1 Introduction

Generative structure-based design has matured quickly: diffusion and autoregressive models can now produce plausible ligands conditioned on a protein pocket (Schneuing et al., 2024; Peng et al., 2022; Guan et al., 2023). The bottleneck has moved downstream. A single campaign can emit thousands of candidates, of which the overwhelming majority cannot be made on a useful timescale, and a further fraction fail liabilities that are cheap to predict and expensive to discover at the bench.

The common response is a filter chain: generate, dock, score, then discard candidates that fail a synthesizability heuristic. This ordering has two costs. First, it throws away generation budget on chemistry that was never makeable, and the surviving set is a residue rather than a design. Second, and more consequentially for the literature, it invites a reporting style in which a docking score is presented as though it were a measurement. Docking scores are not binding affinities (Trott and Olson, 2010), and a redocked co-crystal ligand recovery is a test of pose search, not of discovery.

This paper describes a pipeline built around the opposite ordering (Section 3): synthesizability and liability constraints are evaluated during design, each candidate is emitted with its pose, contact map, liability profile, and a proposed route as a single auditable record, and every prediction carries a confidence label describing the strength of the evidence behind it. We then report what we have and have not established (Sections 4 and 5), and we state the protocol under which a prospective claim could be made.

Contributions.

1. An architecture in which route feasibility and liability screening are generation-time constraints rather than post-hoc filters, and in which the unit of output is an annotated candidate record rather than a ranked score (Table 1).

2. A multi-pose physics-based rescoring rule that selects among docked poses using MM-GBSA support (Genheden and Ryde, 2015), with interaction fingerprints independently confirmed by a second mapping method, so that a contact claim is not the output of a single scorer.

3. An explicit confidence-labelling convention, applied per prediction, whose purpose is to make the boundary between computed and measured quantities legible to a reader who was not present for the campaign.

4. A retrospective pose-recovery case study on LRRK2 with the reference inhibitor MLi-2 (Fell et al., 2015), reported with its protocol and its limits (Section 4).

5. A reporting checklist (Appendix A) that separates in-silico design claims from validated ones, intended to be reusable by other groups reporting generative campaigns.

We state at the outset, and repeat in Section 6, that all results in this paper are in-silico. No compound described here has been synthesized or assayed.

2 Related Work

Structure-based generative models. Diffusion-based and autoregressive pocket-conditioned generators established that chemically valid, pocket-complementary molecules can be sampled directly (Schneuing et al., 2024; Peng et al., 2022; Guan et al., 2023). Their evaluation, however, has centred on geometric and chemical validity rather than on whether a proposed molecule can be made, and increasingly on whether the predicted poses survive independent physical validity checks (Buttenschoen et al., 2024).

Docking, rescoring, and the score-as-measurement problem. Empirical scoring functions remain fast but weakly correlated with affinity (Trott and Olson, 2010); end-to-end learned pose predictors offer an alternative without an explicit scoring function (Torng and Altman, 2019). End-point free-energy methods such as MM-GBSA sit between docking and alchemical free energy in cost and reliability (Genheden and Ryde, 2015). Our use of MM-GBSA is deliberately narrow: it arbitrates between poses of the same ligand, where correlated errors partially cancel, and we do not use it to rank different ligands.

Synthesizability and retrosynthesis. Synthesizability has been formalized as a learned property of proposed molecules (Gao and Coley, 2020) and operationalized through retrosynthetic search over available building blocks. We treat the latter as a feasibility oracle during design: a candidate whose route cannot be closed from commercially available starting materials is deprioritized before it consumes downstream scoring budget.

Liability prediction and its ceiling. Rule-based property filters remain useful summaries (Lipinski et al., 1997; Baell and Holloway, 2010; Bickerton et al., 2012), and QSAR models for absorption, toxicity, and CYP inhibition are standard. Their accuracy outside the applicability domain of their training data is the central caveat, which is why we attach confidence labels to liability predictions rather than treating them as determinations.

3 Pipeline

3.1 Design principles

Three rules govern the implementation.

(i) Constraint before scoring. Any candidate that fails a synthesizability or hard-liability constraint is deprioritized before it consumes docking, rescoring, or liability compute. Generation budget is therefore spent on chemistry that can plausibly be made.

(ii) One candidate, one record. A candidate is never emitted as a bare score. It is emitted with the pose it was scored in, the residue-level contacts that pose makes, its liability profile, a proposed route, and the label of the model that produced each field.

(iii) Nothing is presented as more certain than the evidence behind it. Each prediction carries an explicit confidence label distinguishing, for example, a redocked reference ligand from a prospective proposal, or a prediction inside a model's applicability domain from one outside it. This is a design principle about reporting, not a claim about accuracy.

3.2 Modules

Table 1 lists the stages. The pipeline spans generation through route proposal; module boundaries are deliberately coarse, because the individually interesting question is not how many components exist but whether the record a chemist receives is sufficient to decide what to do next.

3.3 Multi-pose rescoring

For each candidate we retain the top docked poses and compute an end-point binding estimate for each (Genheden and Ryde, 2015). Selection is by best-supported pose, with a stability requirement: a candidate whose poses disagree about the dominant interaction pattern is flagged rather than scored on its most favourable pose. This is a variance-reduction device. It reduces the influence of a single lucky docking run; it does not convert an end-point estimate into a free energy.

3.4 Independently cross-checked contacts

A contact claim is the part of a docket a chemist is most likely to act on, so it is the part most dangerous to produce from one scorer. Each pose's contacts are mapped by two methods, and a contact is reported only where they agree; disagreements are surfaced as uncertainty in the record rather than resolved silently in favour of the more flattering method.

3.5 Synthesizability as a constraint

Route proposal runs inside the design loop (Figure 1, dashed edge). A candidate whose route cannot be closed from commercially available starting materials within the search budget is deprioritized before it reaches the liability screen. We emphasize the scope of this claim: a route proposal establishes that a plausible path exists from purchasable materials. It does not establish yield, impurity profile, cost, or scalability, and it is not a CMC assessment. Those are process-chemistry judgements outside the pipeline.

3.6 Confidence labelling

Labels are assigned per prediction along two axes: *evidence class* (redocked reference ligand, retrospective benchmark, prospective proposal) and *domain position* (inside or outside the training distribution of the model that produced the number). Downstream, a reader can therefore distinguish, without consulting the campaign log, a pose recovered for a ligand

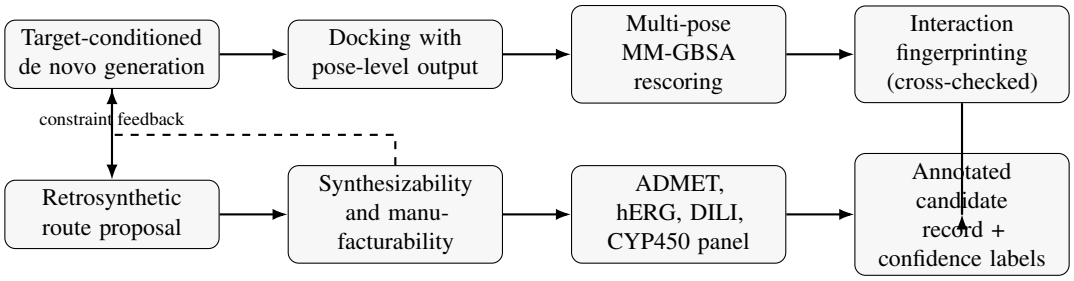

Figure 1: Design-time constraint ordering. Route feasibility and liability screening run in the loop with generation (dashed feedback), rather than as a filter applied to a finished candidate list. The unit of output is an annotated record: structure, pose, contact map, liability profile, proposed route, and per-prediction confidence labels.

Stage	Output and role
Candidate generation	Novel structures generated against the target pocket, not retrieved from a screening library.
Structure-based docking	Pose-level output; the pose, not only a score, is retained for every candidate.
Physics-based rescoring	MM-GBSA across multiple docked poses per candidate; the best-supported pose is selected rather than a single docking run being trusted.
Interaction fingerprinting	Residue-level contacts, independently confirmed by a second interaction-mapping method.
Potency modelling	QSAR potency estimate, labelled with its applicability domain.
ADMET profiling	Per-candidate absorption, distribution, metabolism, excretion and toxicity summary.
Safety and selectivity	Off-target receptor panel; cardiac (hERG) and hepatic (DILI) liability.
Metabolic liability	CYP450 inhibition across 1A2, 2C9, 2C19, 2D6 and 3A4; drug–drug interaction prediction built on that panel.
CNS exposure	Blood–brain barrier penetration model.
Metabolite identification	Predicted metabolites, with reactive and toxic species flagged and bioactivation scored.
Manufacturability	Synthesizability and route-feasibility scoring.
Route proposal	Retrosynthetic routes from commercially available starting materials, per candidate, for chemist adjudication.
Multi-target scoring	Candidates optimized against two or more pockets simultaneously through the same safety panels.
Pose-confidence flagging	Indicates which poses are conformationally reliable, weighted by pocket flexibility so that confidence reflects target-side as well as ligand-side uncertainty.

Table 1: Pipeline stages. Every row is a computation, not a measurement. The confidence convention of Section 3.6 attaches to each.

whose crystal structure is known from a pose proposed for a molecule that has never been made.

4 Retrospective Case Study: LRRK2 and MLi-2

Why this target. LRRK2 is a Parkinson's disease kinase with a well-characterized reference inhibitor, MLi-2 (Fell et al., 2015), whose co-crystal structure supplies a ground-truth pose. Redocking a known ligand tests pose search and scoring against a known answer.

Protocol. The reference ligand is extracted from its co-crystal structure, docked back into the prepared pocket, and evaluated on (i) heavy-atom RMSD to the crystallographic pose, with a 2 Å success threshold, and (ii) recovery of the hinge contacts reported for the complex (GLU1948, ALA1950). Poses are additionally rescored by the multi-pose MM-GBSA rule of Section 3.3.

Reported outcomes and their status. Table 2 lists every quantitative claim associated with this case study alongside its evidentiary status. Two figures in the source material conflict, and one has no retained run log. We report the conflict rather than resolving it by preference; the submission version of this paper will carry only reconciled values, each traceable to a log file.

What this case does and does not show. It shows that the pose-search and rescoring stages can recover a known binding mode for a kinase with a well-defined hinge interaction. It does not show that a novel molecule proposed by the pipeline binds, is selective, is synthesizable in practice, or is therapeutically useful. Redocking a known ligand is a necessary sanity check on the front of the pipeline and is not evidence about its

Quantity	Reported value(s)	Status	What settles it
MLi-2 redock RMSD	Sub-2 Å; separately 2.9 Å → 0.4 Å after rescoring	Conflicting	Retained docking and rescoring logs with seed and receptor preparation
Docking score	-9.5 kcal mol^{-1}	Unverified by artifact	Scoring log from the stated configuration
Campaign runtime	19 min; separately 7.46 h for a nine-target docket on one GPU node	Conflicting	Timestamped job record; the two almost certainly describe different configurations
Model inventory	12+ validated; 18 trained; 24+ modules (different documents)	Conflicting	Frozen manifest with versions and checkpoints
Route-feasibility check	Checked against known easy and hard marketed drugs	Dataset, n and result unrecorded	Re-run with the reference set named in the protocol
Wet-lab validation	None	Absent	Would require synthesis and assay
Prospective discovery	None	Absent	Would require a prospective campaign

Table 2: Quantitative claims for the LRRK2 case study and their evidentiary status at the time of writing. Rows marked conflicting or absent are removed before submission unless reconciled.

back end.

5 What a Prospective Claim Would Require

A single-case redock cannot support a generalization claim. We therefore specify the evaluation we regard as adequate, and we distinguish it from what this draft contains.

Pose recovery at scale. A benchmark of co-crystal complexes, with top-1 and top-5 success at 2 Å, reported per target family, and filtered for physical validity of the predicted poses rather than RMSD alone (Buttenschoen et al., 2024).

Discrimination, not score. Paired active/decoy sets, evaluated by ranking metrics rather than by the magnitude of any single score, to test whether rescoring orders candidates correctly rather than merely producing plausible numbers.

Route feasibility with human adjudication. A named reference set of molecules spanning easy and hard syntheses, with routes judged by chemists against a stated rubric, reporting inter-rater agreement. A synthesizability score without an adjudication protocol is a self-assessment.

Ablations of the design constraints. Generation with and without the synthesizability constraint, to test the central architectural claim: that constraining at design time yields a candidate set that is not merely a filtered subset of the unconstrained one. Multi-pose versus single-pose rescoring, to isolate the variance-reduction contribution. Cross-checked versus single-method contacts, to quantify how often the two methods disagree.

Prospective validation. The only claim that would justify the word *discovery*: candidates selected by the pipeline, synthesized, and assayed, with the selection predating the assay. Until that exists, the appropriate register for this work is methodological.

6 Limitations

No wet-lab validation. No compound described here has been synthesized or tested. Every result is computational.

No affinity claim from docking. Docking and end-point rescoring are used to select poses, not to rank molecules by binding strength. Reported scores should not be read as affinities.

Route proposals are proposals. A closed route from purchasable starting materials demonstrates plausibility, not yield, purity, cost, or scalability.

Single-case retrospective. The case study covers one target and one reference ligand, chosen because a ground-truth pose exists. It is a sanity check, not a benchmark.

Applicability domain. Learned liability and potency models are reliable only within the chemistry they were trained on; predictions outside that domain carry the corresponding label and should be treated as hypotheses.

Compute reproducibility. Configurations reported in earlier internal documents disagree on runtime; this draft therefore makes no performance claim until the configurations are reconciled and logged.

7 Reproducibility and Availability

The submission version will report, for every table entry, the command, seed, receptor preparation, model checkpoint identifier, and hardware used to produce it, and will state which

artifacts are released with the paper. Where an artifact cannot be released, the reason will be given rather than the row silently omitted.

8 Conclusion

Treating synthesizability as a constraint at design time, and attaching an explicit evidence label to every prediction, changes what a generative campaign hands to a chemist: not a ranked list of scores, but a set of candidates whose provenance is legible. We have described such a pipeline, set out a retrospective case study with its limits stated, and specified the evaluation that would be needed before any prospective claim could be made. The most useful contribution at this stage may be the discipline of the reporting rather than the numbers: a ranked docket and a validated hit are different objects, and conflating them is the failure mode this work is built to prevent.

A Reporting checklist

Each item should be answerable from the paper or its supplement without contacting the authors.

1. Is every number in the results traceable to a named artifact (log, checkpoint, dataset version)?
2. Are conflicting internal figures reconciled, with the discarded value and the reason stated?
3. Is the distinction between redocked reference ligand, retrospective benchmark, and prospective proposal explicit at the point of each claim?
4. Is the applicability domain of each learned model stated, and are out-of-domain predictions labelled?
5. Are route proposals accompanied by an adjudication protocol and by agreement statistics?
6. Is hardware, runtime, seed and checkpoint reported for every experiment?
7. Is it stated, in the abstract and not only in a limitations section, that no wet-lab validation exists?
8. Are data and code availability statements specific about what is released, and why anything withheld is withheld?
9. Has all internal commercial material been excluded from the submission, including partner names, target identities under agreement, and non-public commercial strategy?

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